TEXORA · LECTURE SHEET

Carbonyl Compounds

Aldehydes and ketones: the polar C=O group and its nucleophilic-addition chemistry, preparations, the tests that separate aldehydes from ketones (Tollens, Fehling, 2,4-DNP), reductions, aldol condensation, Cannizzaro, and the haloform reaction.

CHE-1205 · CHEMISTRY-II · TOPIC 3 · LEVEL 1 TERM 2

1. The carbonyl group

Aldehydes (R–CHO) and ketones (R–CO–R′) share the **carbonyl group** C=O: planar, sp², and strongly polarised (C δ+, O δ−). That polarisation writes the whole chemistry: **nucleophiles attack the carbon**, electrophiles the oxygen. Aldehydes are the more reactive class — they carry only one electron-donating alkyl group (ketones have two) and present less steric crowding; formaldehyde, with none, is the most reactive of all. Small members are water-soluble through hydrogen bonding to the oxygen; boiling points sit between alkanes and alcohols of similar mass.

2. Preparation

ROUTE	NOTES
Oxidation of alcohols	1° alcohol → aldehyde (mild oxidant, e.g. PCC, stops there; strong oxidant continues to the acid); 2° alcohol → ketone
Ozonolysis of alkenes	C=C cleaved to carbonyls; each alkene carbon becomes a carbonyl carbon
Rosenmund reduction	Acyl chloride + H_2 over Pd/$BaSO_4$ (poisoned catalyst) stops at the aldehyde
Stephen reaction	Nitrile + $SnCl_2$/HCl, then hydrolysis → aldehyde
Hydration of alkynes	$HgSO_4$/H_2SO_4 adds water Markovnikov-style: terminal alkynes give methyl ketones; ethyne alone gives acetaldehyde
Friedel–Crafts acylation	Benzene + RCOCl/$AlCl_3$ → aromatic ketones (acetophenone and family)

3. Nucleophilic addition — the signature reaction

- **HCN** → cyanohydrins (extends the chain by one carbon; gateway to α-hydroxy acids).

- **$NaHSO_3$** → crystalline bisulphite adducts from aldehydes and methyl ketones only — used to purify and identify liquid carbonyls; the addition reverses with acid or alkali.
- **Grignard reagents** → alcohols after work-up (formaldehyde → 1°, aldehydes → 2°, ketones → 3°): the standard carbon-skeleton builder.
- **Alcohols** → hemiacetals, then **acetals** with a second equivalent (acid catalyst) — protecting groups for the carbonyl.
- **Ammonia derivatives** (NH_2OH, NH_2NH_2, 2,4-dinitrophenylhydrazine) → oximes, hydrazones and 2,4-DNP derivatives by addition–elimination. The 2,4-DNP product is a bright orange-red crystalline solid: **Brady's test**, positive for both aldehydes and ketones, with sharp melting points used for identification.

4. Oxidation — where aldehydes and ketones part company

The aldehyde's hydrogen makes it a reducing agent; ketones resist mild oxidants. Three classical tests exploit this:

TEST	ALDEHYDES	KETONES
Tollens (ammoniacal $AgNO_3$, warm)	Silver mirror (aromatic aldehydes included)	No reaction
Fehling's / Benedict's	Brick-red Cu_2O precipitate — aliphatic aldehydes; benzaldehyde fails	No reaction
Acidified $KMnO_4$	Decolourised; aldehyde → carboxylic acid	Only under vigorous conditions (C–C cleavage)

Reductions run the other way: $NaBH_4$ or $LiAlH_4$ returns carbonyls to alcohols; **Clemmensen** (Zn–Hg/conc. HCl) and **Wolff–Kishner** (NH_2NH_2, KOH, heat) both take C=O all the way to CH_2 — choose Clemmensen for acid-stable molecules, Wolff–Kishner for base-stable ones.

5. Reactions of the α-hydrogen

Hydrogens on the carbon next to C=O (α-hydrogens) are weakly acidic, and their chemistry defines the class's condensation reactions:

- **Aldol condensation:** with dilute NaOH, two molecules of an α-H carbonyl combine: acetaldehyde → 3-hydroxybutanal (the "aldol"), which dehydrates on heating to crotonaldehyde (but-2-enal), an α,β-unsaturated carbonyl. Crossed aldols between two different partners are useful when one partner has no α-H.

- **Cannizzaro reaction:** carbonyls *without* α-H (benzaldehyde, formaldehyde) cannot aldol; with concentrated KOH they disproportionate instead — one molecule oxidised, one reduced: 2 $C_6H_5CHO \rightarrow C_6H_5CH_2OH + C_6H_5COOK$.
- **Haloform reaction:** methyl ketones ($CH_3CO–$) with I_2/NaOH give yellow iodoform, CHI_3, plus the carboxylate of the rest of the molecule (one carbon shorter). Acetaldehyde is the only aldehyde with a $CH_3CO–$ unit, hence the only aldehyde answering the iodoform test.

6. Key points

- C=O is polar and planar; nucleophilic addition is the characteristic reaction; reactivity aldehyde > ketone, formaldehyde highest.
- Tollens separates all aldehydes from ketones; Fehling's only aliphatic aldehydes; 2,4-DNP flags any carbonyl.
- Clemmensen (acidic) and Wolff–Kishner (basic) both reduce C=O → CH_2.
- α-H present → aldol; α-H absent + conc. alkali → Cannizzaro; $CH_3CO–$ present → haloform/iodoform.

7. Practice

Answer in your own words; brief model answers follow.

1. Propanal and propanone are isomers. Give two chemical tests that distinguish them, with observations.
2. Acetaldehyde is treated with dilute NaOH, then the mixture is warmed. Name both products in sequence and the reaction type.
3. Benzaldehyde with concentrated KOH gives benzyl alcohol and potassium benzoate. Name the reaction, and explain why benzaldehyde cannot undergo aldol condensation instead.
4. Write the outcome when acetone is warmed with I_2/NaOH, and name the yellow solid. Which single aldehyde would behave the same way?
5. Starting from benzoyl chloride, how is benzaldehyde prepared? Name the reaction and the device that prevents over-reduction.
6. Acetaldehyde adds HCN faster than acetone. Give the two reasons (electronic and steric).

Answer key

1. Tollens' reagent: propanal deposits a silver mirror; propanone does nothing. Fehling's solution: propanal gives a brick-red Cu_2O precipitate on warming; propanone leaves the blue solution

unchanged. (Both give orange precipitates with 2,4-DNP — that test detects the carbonyl but cannot separate the pair.)

2. First 3-hydroxybutanal (aldol addition product); on warming it dehydrates to but-2-enal (crotonaldehyde). The sequence is aldol condensation.
3. The Cannizzaro reaction (self oxidation–reduction/disproportionation). Aldol condensation requires an α-hydrogen to form the attacking enolate; benzaldehyde's carbonyl carbon is attached directly to the ring — no α-carbon, no α-hydrogen, no aldol.
4. Yellow iodoform, CHI_3, precipitates, leaving sodium acetate in solution: $CH_3COCH_3 + 3\ I_2 + 4\ NaOH \rightarrow CHI_3\downarrow + CH_3COONa + 3\ NaI + 3\ H_2O$. The aldehyde is acetaldehyde — the only one containing the $CH_3CO–$ group.
5. Rosenmund reduction: H_2 passed over Pd supported on $BaSO_4$, the catalyst deliberately poisoned (e.g. with sulphur/quinoline) so reduction halts at the aldehyde instead of continuing to benzyl alcohol.
6. Electronic: acetone's two methyl groups donate electron density (+I), shrinking the carbonyl carbon's δ+ and its appetite for the nucleophile; acetaldehyde has one methyl and one hydrogen. Steric: two methyls crowd the approach and the tetrahedral product; acetaldehyde presents a small hydrogen on one side. Both effects rank aldehydes above ketones toward every nucleophile.
